Algebra Essentials

Section 1: Linear Equations & Distribution

1 If $3(2x - 4) = 4x + 6$, what is the value of x?

a. 9

b. 6

c. 8

d. 10

2 If $\frac{x}{3} + \frac{x}{4} = 14$, what is the value of x?

a. 24

b. 21

c. 28

d. 18

3 If $2x + y = 17$ and $x - y = 4$, what is the value of xy?

a. 21

b. 18

c. 24

d. 27

Section 2: Difference of Squares & Algebraic Identities

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4 If $x^2 - y^2 = 48$ and $x - y = 6$, what is the value of $x + y$?

a. 8

b. 6

c. 12

d. 14

5 If $x + y = 10$ and $xy = 21$, what is the value of $x^2 + y^2$?

a. 58

b. 62

c. 54

d. 79

6 Evaluate the numerical expression: $101^2 - 99^2$.

a. 400

b. 200

c. 800

d. 300

Section 3: Inequalities & Absolute Values

7 What is the solution set of $|2x - 5| = 9$?

a. $\{7, -2\}$

b. $\{7, 2\}$

c. $\{-7, 2\}$

d. $\{5, -4\}$

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8 If $-3x + 7 \leq -5$, what is the range of x?

a. $x \geq 4$

b. $x \leq 4$

c. $x \geq -4$

d. $x \leq -4$

9 If $x < y < 0$, which of the following expressions must be positive?

a. x^2y^2

b. $x - y$

c. $x + y$

d. $\frac{x}{y^3}$

Section 4: Advanced Systems & Substitution Tricks

10 If $\frac{1}{x} + \frac{1}{y} = \frac{5}{6}$ and $xy = 12$, what is the value of $x + y$?

a. 10

b. 12

c. 8

d. 15

11 If $a + b + c = 18$ and $a + b = 11$, while $b + c = 13$, what is the value of b?

a. 6

b. 5

c. 7

d. 8

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12 If $x^2 + 6x + 9 = 25$ and $x > 0$, what is the value of x?

a. 2

b. 3

c. 4

d. 5

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